Digital Design Corporation / Advanced Network Devices (IP mass-notification)

Digital Design Corporation / Advanced Network Devices (IP mass-notification) — Firmware Attack Surface & Detection Brief

Scope: CVE and public-advisory (OSINT) research only — **no firmware or software has been acquired or extracted for this vendor.** Every finding below comes from public vulnerability disclosures (NVD, CISA ICS advisories, vendor PSIRT) and product documentation. CVE IDs listed in the Confirmed table were each independently re-verified against NVD/CISA/vendor records before inclusion; candidate IDs that could not be confirmed were dropped, not guessed.

> **Lineage / cross-reference.** The census string 'Digital Design Corporation' resolves to Advanced Network Devices ('AND'); it is NOT the same company as 'Digital Systems Corp' (Galaxy Control Systems), which has its own brief.

Identity

Field	Value
Census / OUI string	`Digital Design Corporation`
Operating company	Advanced Network Devices, Inc. (brand "AND", anetd.com), the networked-endpoint product arm co-located with and sharing the IEEE OUI of Digital Design Corporation (digidescorp.com). Both are Donahoe-family firms in the same Arlington Heights, IL facility; DDC is the FPGA/embedded engineering parent and OUI registrant of record, AND is the commercial OT product brand under which the census devices actually ship.
OT devices (ARNG census)	160 across 11 states
Genuine OT vendor?	Yes
Purdue placement	Level 1/2 field edge devices of a building mass-notification / public-address / life-safety OT system. The endpoints (speakers, displays, strobes, intercoms, GPIO relays) are the Level 1 actuators/sensors; the IPClockWise / InformaCast / Syn-Apps / BellCommander servers and SIP PBX that drive them are the Level 2-3 supervisory layer. On a typical flat Guard facility network they sit in a building-systems/physical-security VLAN alongside cameras, access control and BAS, which is where the census found them.

Identity note. The census string is the OUI owner, not the product brand. OUI 00:11:90 (IEEE MA-L, registered 2004-06-15, last updated 2015-11-17) is registered to "Digital Design Corporation, 3820 Ventura Dr, Arlington Heights, IL 60004" (confirmed via maclookup.app). DDC (digidescorp.com) is an FPGA/ASIC/board design house whose own branded products are VAADR military/airborne video-and-data recorders and image-enhancement platforms — not the devices one would expect to see 160x across Guard facilities. DDC's corporate overview, however, lists an "Advanced Network Products & Solutions" line, and its sister/spinoff company Advanced Network Devices, Inc. (anetd.com, founded 2004 by Brian Donahoe; DDC founder/CEO is Debbie Donahoe) sits at 3860 N Ventura Dr in the same park, shares the same fax line (847-359-5418), and ships its IP endpoints under the shared OUI 00:11:90. Therefore devices carrying 00:11:90 on an OT/physical-security VLAN are almost certainly AND IP speakers/displays/clocks/intercoms, not DDC avionics recorders. Map the census entry to the AND / anetd.com product line.

Product lines & OT relevance

Genuine OT / life-safety building system. AND (under OUI 00:11:90) makes IP-based mass-notification, emergency paging, intercom, bell-schedule and clock/display endpoints — PoE devices that live on the facility network and are exactly the kind of gear that appears in large counts (160 across 11 states is consistent with mass-notification/giant-voice, paging, intercom and clock coverage across many Guard buildings). They belong in the same physical-security/building-automation census bucket as the other vendors on the ARNG exposure page (Genetec, Hanwha Vision, Motorola Solutions, Trane, OSI Security Devices). This is NOT an OUI-artifact false positive and NOT an IT vendor misfiled as OT: the endpoints themselves are the networked OT assets. One nuance worth flagging to the SOC: the built-in microphone + two-way "monitoring"/paging-supervision on the speaker line makes these also an audio-surveillance exposure, not just an availability/notification concern.

- IP speakers (PoE/PoE+) incl. IPSWD (speaker + LED display), IPSWS surface/ceiling mount, indoor/outdoor models; built-in microphone for two-way talkback/intercom and 'paging supervision'/monitoring
- IP clocks / LED message displays (IPCS, IPCSL large-format, IPDisplay; NTP-synced, scrolling text, RSS/weather/alert feeds, double-sided models)
- IP visual-notification strobes/flashers and IP call/panic buttons (GPIO: 2 inputs, 1 output, activatable by phone or programmatically)
- IP intercom / classroom-to-office intercom and zone/overhead paging endpoints
- PoE hubs/injectors and mounting accessories
- Management/software: AND IPClockWise / 'Clockwise Campus' (bell schedules, alarms, events, sound library) and 'Vigilar' threat-detection; interoperates with Singlewire InformaCast, Syn-Apps SA-Announce, IPCelerate IPSession, MessageNet, AcroVista BellCommander, Cisco, and any SIP PBX (Asterisk/3CX/ShoreTel)
- (Parent DDC branded line, NOT the census devices: VAADR military/airborne video-and-data recorders, real-time image-enhancement platforms, AVB/TSN FPGA IP cores)

Attack surface

- Protocol set per AND's own IPSWD datasheet: SIP, HTTP (cleartext web admin; no HTTPS advertised on these endpoints), SLP, TFTP, NTP, SNMPv1, DHCP, IGMP, ICMP, mDNS/mDNS-SD (Bonjour),

IPv4/TCP/UDP, plus full multicast + broadcast audio/text.

- Untrusted zero-touch provisioning is the headline vector: on boot the device auto-discovers a config file via DHCP options or SLP, then downloads BOTH configuration AND firmware over cleartext TFTP/HTTP from a server supplied by (1) SLP, (2) DHCP option 150 TFTP, (3) DHCP option 66 TFTP, or (4) DHCP option 72 HTTP (dir '/ipspeaker/'). A rogue DHCP server or spoofed SLP responder on the L2 segment can redirect endpoints to attacker-controlled firmware/config — effectively an unauthenticated firmware-implant path.
- SIP credentials (extension id, password, digest_username) are stored in cleartext XML config files (e.g. AllSpeakers.cfg) delivered over TFTP/HTTP; AND's own guide example uses password="password123". Intercepting or pulling the provisioning file yields PBX-registerable SIP creds and a pivot/toll-fraud path.
- Built-in microphone with two-way talkback, hands-free 'monitoring' and 'paging supervision' on the speaker line = live audio eavesdropping risk if an endpoint or its control server is compromised — a counterintelligence concern at a Guard facility.
- Web server is 'field upgradable' (firmware uploaded via the on-box web UI). Public datasheets/guides do not document any default web-admin credential or mandatory auth; historically this class of endpoint shipped with weak or no enforced web authentication. Treat the web interface as unauthenticated-until-proven and verify on a sample; do NOT assume a hardened login.
- SNMPv1 only (cleartext, community-string based — default/guessable community strings typical) for status/management.
- SLP (udp/427) and mDNS/Bonjour (udp/5353) broadcast discovery expose device name/IP/firmware version for easy enumeration; devices are also reachable by Bonjour hostname (IPSpeaker.local) and IPv6 link-local.
- Network-addressable GPIO (2 in / 1 out) 'activatable via telephone or programmatically' allows remote actuation of attached relays/strobes/contacts.

Confirmed CVEs / advisories

No CVE or advisory was confirmed for this vendor in this pass. This is recorded as an **absence of information, not a clean bill of health** — for a vendor of this fleet size the likelier explanations are limited public research attention or a vendor that does not publish advisories where researchers find them. A direct CISA/vendor-PSIRT pass and, ultimately, firmware acquisition are the ways to close it.

Detection guidance

- Asset-ID: sweep OT/building-systems VLANs for OUI 00:11:90 (IEEE MA-L); expect clusters of IP speakers/clocks/displays. Reconcile the 160-device census against the authorized IPClockWise/InformaCast inventory.
- Guard the provisioning trust chain: enable DHCP snooping and alert on any rogue DHCP server and on DHCP options 66/150 (TFTP) and 72 (HTTP '/ipspeaker/') offered by anything other than the sanctioned provisioning host; monitor SLP (udp/427) responses on the endpoint VLAN.
- Baseline and alert on TFTP (udp/69) and HTTP (tcp/80) config/firmware transfers to/from OUI 00:11:90 endpoints — permit only the one authorized IPClockWise/provisioning server; alert on firmware-upload POSTs to endpoint web UIs from any other host.

- SIP (5060/udp,tcp): permit registration only to the sanctioned PBX/registrar; alert on SIP from these endpoints to any external or unknown registrar (exfil/toll-fraud/pivot).
- Multicast: baseline expected IGMP joins / RTP multicast audio groups and the InformaCast SLP multicast group 239.255.255.253; alert on new or external multicast senders.
- SNMPv1 (udp/161): detect default/guessable community strings and unauthorized SNMP from off-VLAN; disable SNMPv1 if unused. Watch mDNS (udp/5353) and SLP (udp/427) for discovery/enumeration chatter.
- Log sources to onboard: switch DHCP-snooping logs, NetFlow/IPFIX on the building-systems VLAN, SIP registrar/PBX logs (Cisco CUCM, Asterisk, 3CX, ShoreTel), and the IPClockWise / InformaCast (Singlewire) / Syn-Apps SA-Announce / BellCommander / MessageNet server logs plus the SNMP trap receiver.
- Allowlist/segmentation to enforce and monitor: pin each endpoint to its provisioning host (TFTP/HTTP) + NTP server + SIP registrar + its multicast groups only; deny internet egress, deny inter-VLAN except to sanctioned servers, and disable zero-touch provisioning after commissioning where the firmware allows.

Acquisition status

We currently hold NONE for this vendor. Firmware is gated: images and the IPClockWise application live behind the AND Customer Portal login at anetd.com/user-support/portal-login/ (account required; the device status page even tells operators to log in there to check for the latest firmware). No open/direct firmware download was found. Public, ungated material is limited to datasheets, install guides and the SIP-config application note (anetd.com technical-resources, and mirrors on fullcompass.com). Two realistic acquisition paths, both requiring authorization: (1) request images through the AND portal / support (tech@anetd.com, 847-463-2237); (2) because firmware is pushed in-band to devices over cleartext TFTP/HTTP during provisioning, a sample can be captured on-wire from the authorized provisioning server on the Guard network with proper authorization. Net: vendor-gated, obtainable only via portal account or on-net capture.

Notes

No CVE or ICS-CERT/CISA advisory was found for this vendor in this research, so cve_claims is intentionally empty (none fabricated). Evidence checked: NVD REST API keywordSearch=anetd (exact) returned totalResults:0; keywordSearch="Advanced Network Devices" returned only generic 'network devices' hits for Siemens SIMATIC and Juniper — those are NOT this vendor and do not apply; no CISA ICS advisory, ZDI, or vendor security bulletin surfaced for AND/anetd/Digital Design Corporation. Absence of a published CVE is not assurance of safety: the cleartext TFTP/HTTP provisioning-and-firmware trust model, SNMPv1, cleartext-stored SIP creds, and undocumented web-admin auth posture are high-value review targets and likely pre-authentication/weak-auth issues if these endpoints were ever formally tested. Recommended next step for the SOC: pull one device's firmware (portal or on-net capture) to confirm the web-admin default-credential/auth state and whether firmware/config downloads are signed — none of the public documentation indicates image signing. Primary sources used: maclookup.app OUI 00:11:90; digidesco.com / digidescorp.com (overview PDF); anetd.com (about-us, IPSWD datasheet, IP Device Installation Guide, AN060 SIP config app-note); fullcompass.com datasheet mirrors; NVD REST API.

Status

Item	State
Firmware/software acquired	**No** — OSINT/advisory only
Confirmed CVEs	0 (independently re-verified)
Behavioural baseline	None — no samples held
Next step	CISA/PSIRT deep pass; attempt firmware acquisition

Generated 2026-10-26 from the ot-vendor-briefs workflow (12 researchers + adversarial CVE verification). Device counts from the 2026-10-22 ARNG OT census.